

Managed Copilot

Run GitHub Copilot agents in environments you own: on your laptop, or in your Azure subscription.

A tour of what we built · stock GitHub Copilot SDK 1.0.26 and runtime 1.0.103 · Aspire · Azure Container Apps

THE IDEA

Same Copilot SDK. Your environment.

```
import { approveAll } from "@github/copilot-sdk";
import { ManagedCopilot } from "@managed-copilot/sdk";

const managed = await ManagedCopilot.connect();    // your platform
const client = await managed.client("chat");  // a stock CopilotClient

const session = await client.createSession({
  model: "claude-sonnet-5", onPermissionRequest: approveAll,
});
const reply = await session.sendAndWait({ prompt: "Summarize what changed" });
```

- Genuine **`CopilotClient`** and **`CopilotSession`** objects
- Two setup lines point it at your platform
- The published SDK and runtime, unmodified
- The platform decides where agents run and what they reach

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ENVIRONMENTS

Three kinds of environments

Shared

A lightweight chat agent on a shared runtime. Its tools run inside your app.

`type: "shared"`

Isolated

Every session gets its own sandbox: a workspace, a shell, your packages, and network rules you control.

`type: "isolated"`

Promotable

Starts cheap on the shared runtime, then moves into its own sandbox when it needs one.

`type: "promotable"`

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// apphost.mts: declared once, picked by name
await addManagedCopilot(builder, "copilot", { environments: {
  chat:   { type: "shared", network: ["learn.microsoft.com"] },
  coder:  { type: "isolated", size: "medium" },
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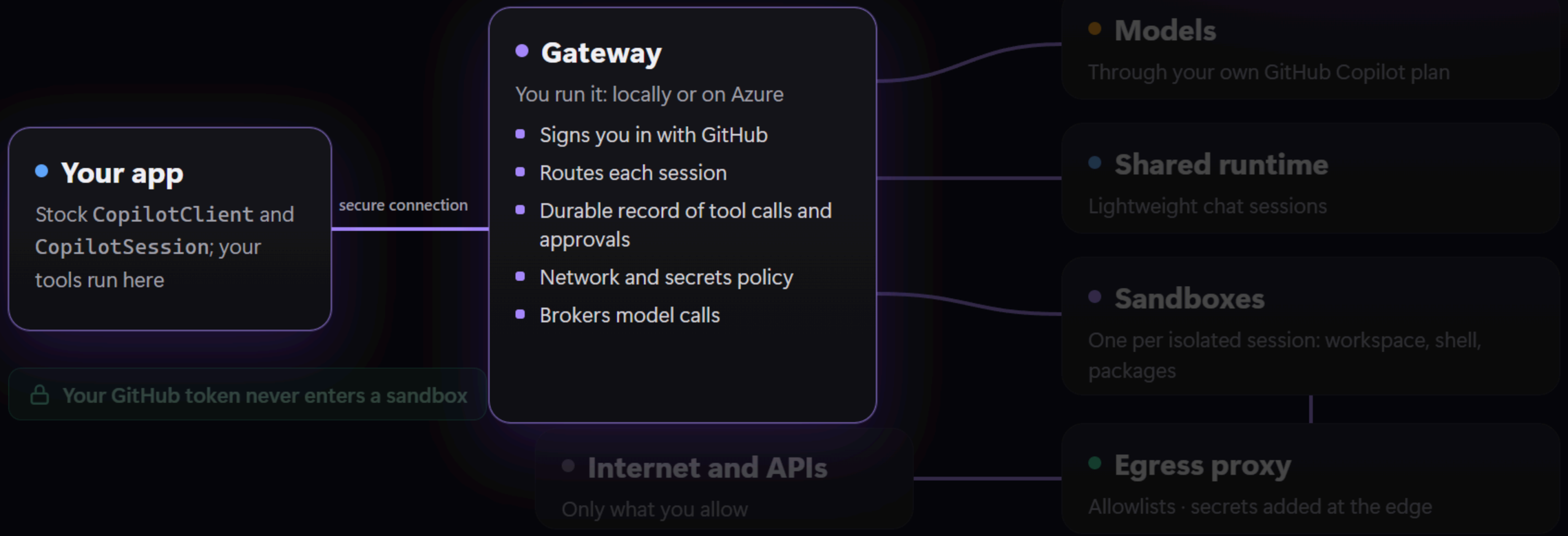
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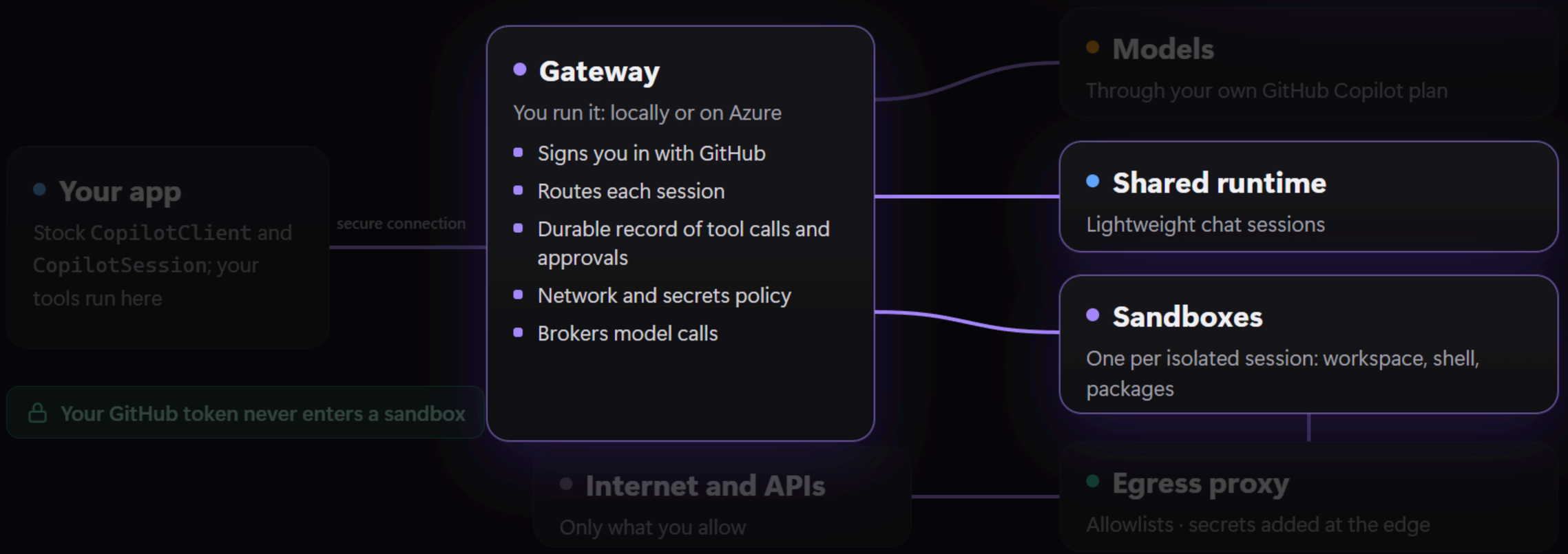
HOW IT WORKS

How the pieces fit together



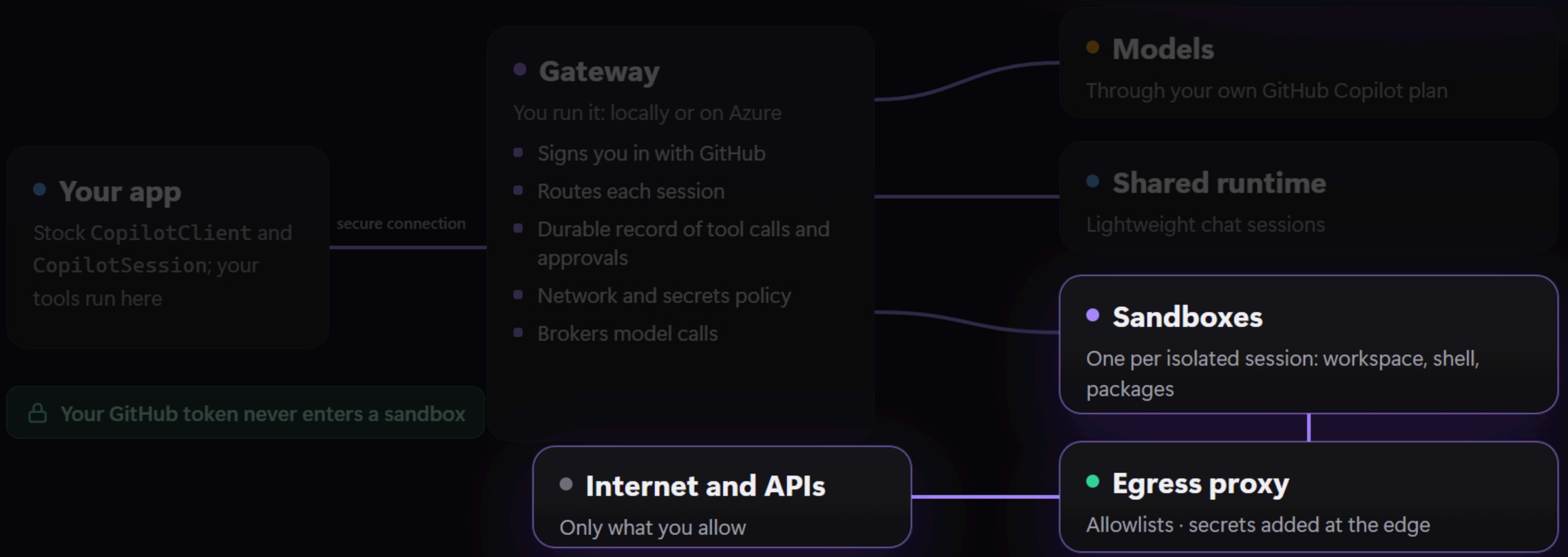
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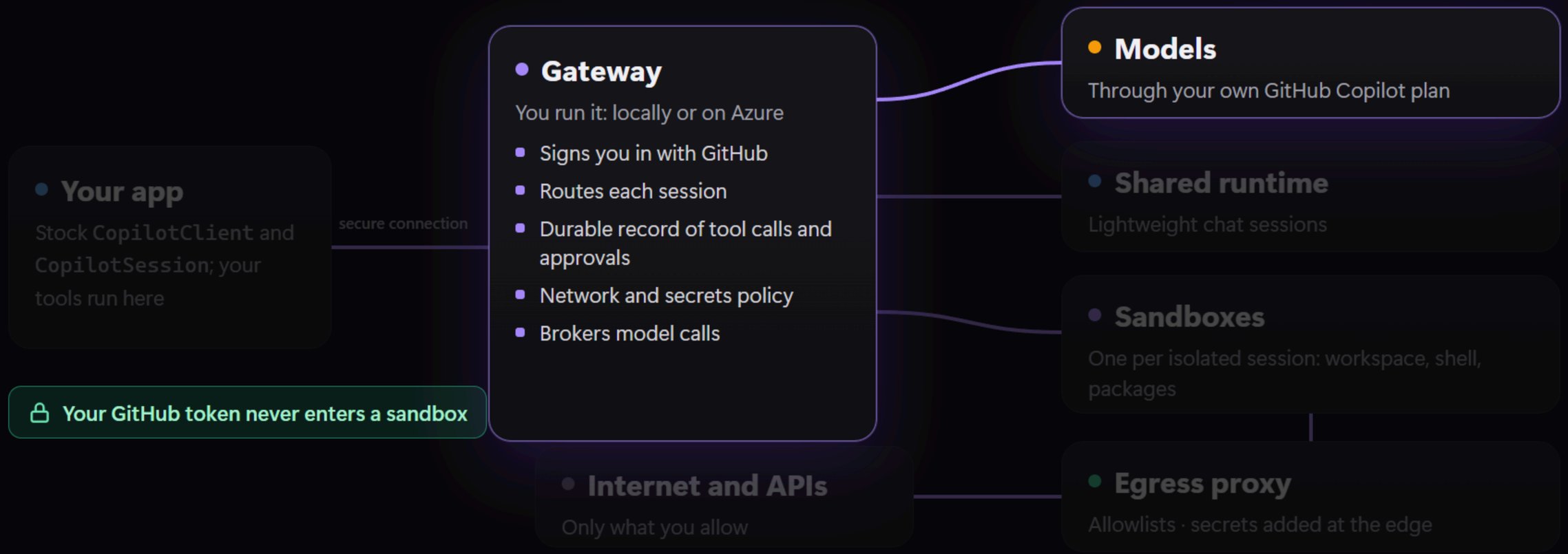
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RUN IT

One command: `aspire run`

aspire run · resources

RESOURCE	WHAT IT IS	STATE	URL
copilot	Gateway, shared runtime, sandboxes, egress proxy	● Running	localhost:7400
software-factory	Demo 1: AI software factory	● Running	localhost:7480
assistant	Demo 2: personal assistant	● Running	localhost:7490
quickstart	Every spec example as a check	Start from the dashboard	–

VERIFIED

Every spec example is an automated check

quickstart · real platform, real models

```
✓ 01 The idea in one snippet
✓ 02 Shared: a chat agent with your tools
✓ 03 Isolated: a coding agent in a sandbox
✓ 04 Set up the sandbox
✓ 05 Get results out
✓ 06 Control network access
✓ 07 Keep secrets out of the sandbox
✓ 08 Promotable: start cheap, move later
✓ 09 Come back later
✓ 10 Offline work: webhook
✓ 11 Offline work: polling
✓ 12 Run on a schedule
✓ 13 Open a session in Copilot CLI

13 passed
```

DEMO 1

AI software factory

Describe the software. A team of Copilot agents plans, builds, reviews, and releases it.

DEMO 2

Personal assistant

A bot you chat with, connected to mail and calendar, that keeps working while you're away.

WHILE YOU'RE AWAY

The tab is closed. The work continues.

+0:00	You ask for a check-in in two minutes, then close the tab The platform keeps the schedule; it runs as you after a one-time sign-in
+2:00	The platform runs the request with no browser open The agent needs your calendar, so the platform calls the assistant's server through its webhook
Later	You come back to the answer The calendar lookup is marked "Handled while you were away"

ON AZURE

Same AppHost: `aspire deploy`

- **Azure Container Apps**
  - Gateway
  - Software factory
  - Assistant
  - Egress proxy
- **Container Apps Sandboxes**

  One sandbox per isolated session, with data disks that survive compute replacement
- **Azure Files**

  The platform's state, with a single-writer lease

Apps are restricted to your allowed IP ranges, and the gateway accepts only the GitHub users or orgs you name.

RECAP

What we built

- ✓ Genuine Copilot SDK clients and sessions, unmodified
- ✓ Sandboxes with setup, files, and artifacts
- ✓ Offline work and schedules that run as you
- ✓ Two demos, end to end with real models
- ✓ Shared, isolated, and promotable environments
- ✓ Network allowlists and secrets added at the edge
- ✓ Copilot CLI on the same live session
- ✓ Runs locally with Aspire and on Azure

TRY IT

Run it yourself

terminal

```
> gh auth login
> pnpm install
> pnpm build
> aspire run
```

- Software factory: `localhost:7480`
- Assistant: `localhost:7490`
- Start with `README.md` and `demos/quickstart`
- Deploy with `docs/deployment.md`